

R MIEF Skills Workshop

Session 2

September 19, 2025

Today

- Learn how to:
 - Create a scatter plot, density plot, and bar chart using the `ggplot2` package
 - Create flexible and easy-to-read tables of any dataset using the `gt` package
 - Create simple academic-standard regression output tables using the `stargazer` package
- Practice the above!

Data Visualization — Descriptive Statistics — Plots

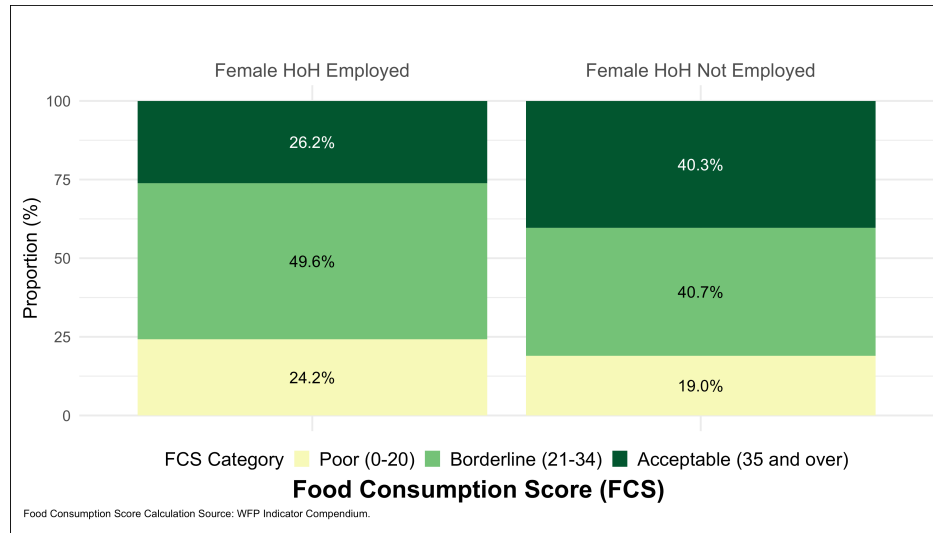
Descriptive Stats Plots

`ggplot2` is the gold standard in data visualization in data work. It's one of the main reason that people use R over other programming languages.

Very simple syntax and allows you to add elements very easily.

You can use `ggplot2` to create any type of plot you can think of.

I've included a lot of links at the end of these slides to explore the possibilities of `ggplot2` further. Strongly recommend you use them or at least save them somewhere.



The Magic of `ggplot2`

Using `ggplot2` to create plots is great because the **structure** it sets up makes plot creation intuitive.

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(  
    mapping = aes(<MAPPINGS>),  
    stat = <STAT>,  
    position = <POSITION>  
  ) +  
  <SCALE_FUNCTION> +  
  <FACET_FUNCTION> +  
  <THEME_FUNCTION>
```

1. **Data**: The data that you want to visualize
2. **Layers**: `geom_` and `stat_` → The geometric shapes and statistical summaries representing the data
3. **Aesthetics**: `aes()` → Aesthetic mappings of the geometric and statistical objects
4. **Scales**: `scale_` → Maps between the data and the aesthetic dimensions
5. **Facets**: `facet_` → The arrangement of the data into a grid of plots
6. **Visual themes**: `theme()` and `theme_` → The overall visual defaults of a plot

Scatter Plot — Step-by-Step

Dataset

Convert to Plot

Add Something

Start with a dataset you want to visualize

```
head(mtcars)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
##	Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
##	Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
##	Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
##	Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

Scatter Plot — Step-by-Step

Dataset

Convert to Plot

Add Something

```
ggplot(mtcars)
```

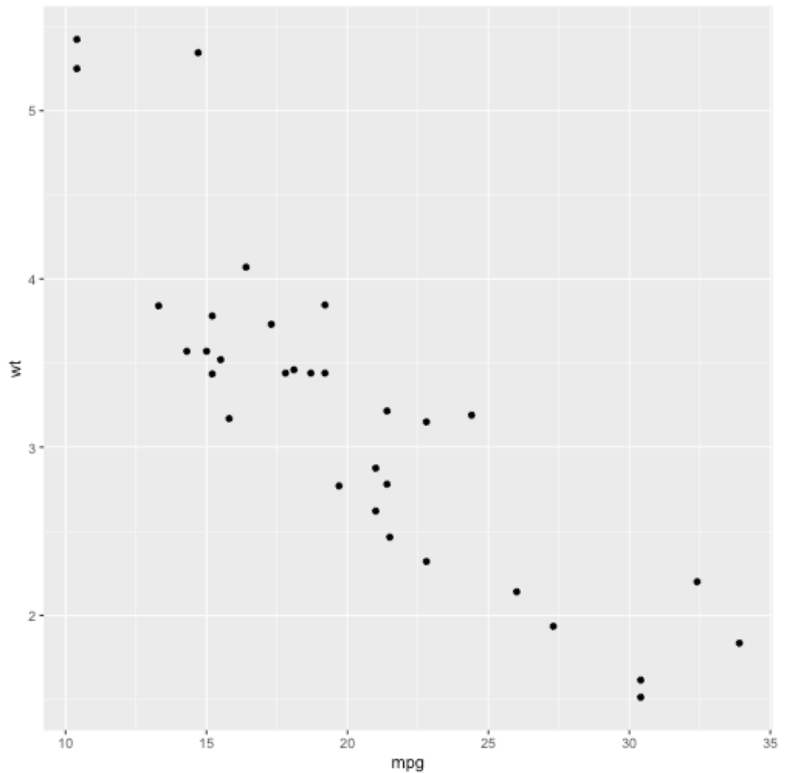
Scatter Plot — Step-by-Step

Dataset

Convert to Plot

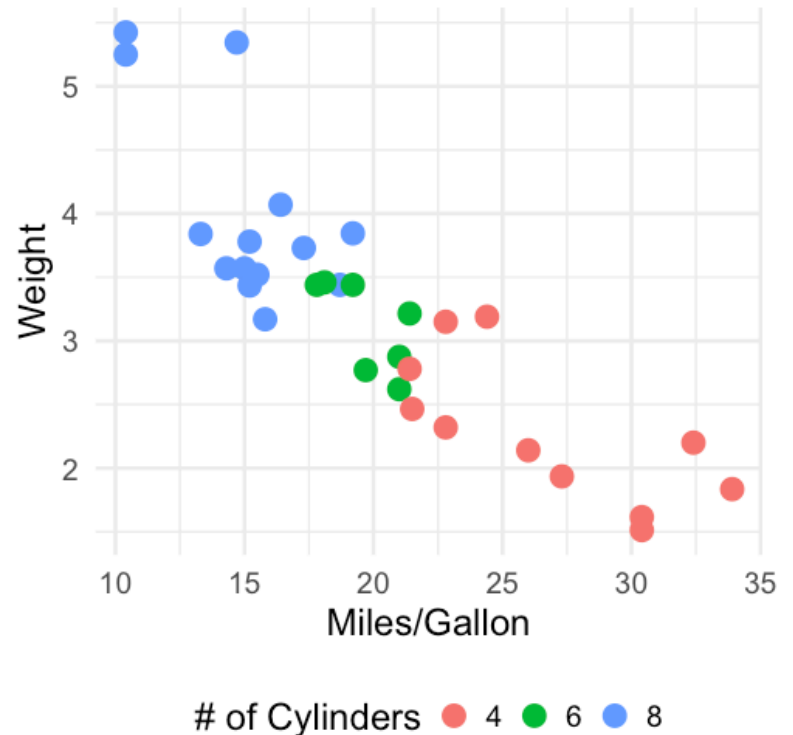
Add Something

```
ggplot(mtcars) +  
  geom_point(  
    aes(x = mpg, y = wt)  
  )
```



Scatter Plot — Make It Better

```
ggplot(mtcars) +  
  geom_point(  
    aes(  
      x = mpg, y = wt,  
      color = factor(cyl)  
    ),  
    size = 6  
  ) +  
  xlab("Miles/Gallon") +  
  ylab("Weight") +  
  scale_color_discrete(  
    name = "# of Cylinders"  
  ) +  
  theme_minimal(base_size = 24) +  
  theme(  
    legend.position = "bottom"  
  )
```



Bar Plot — Step-by-Step

Dataset

Convert to Plot

Add Something

Fix Class Issue

Start with a dataset you want to visualize

```
mtcars_summary <- mtcars %>%  
  group_by(cyl) %>%  
  summarize(  
    mpg = mean(mpg)  
  ) %>%  
  ungroup()  
  
mtcars_summary
```

```
## # A tibble: 3 × 2  
##   cyl  mpg  
##   <dbl> <dbl>  
## 1     4  26.7  
## 2     6  19.7  
## 3     8  15.1
```

Bar Plot — Step-by-Step

Dataset

Convert to Plot

Add Something

Fix Class Issue

```
ggplot(mtcars_summary)
```

Bar Plot — Step-by-Step

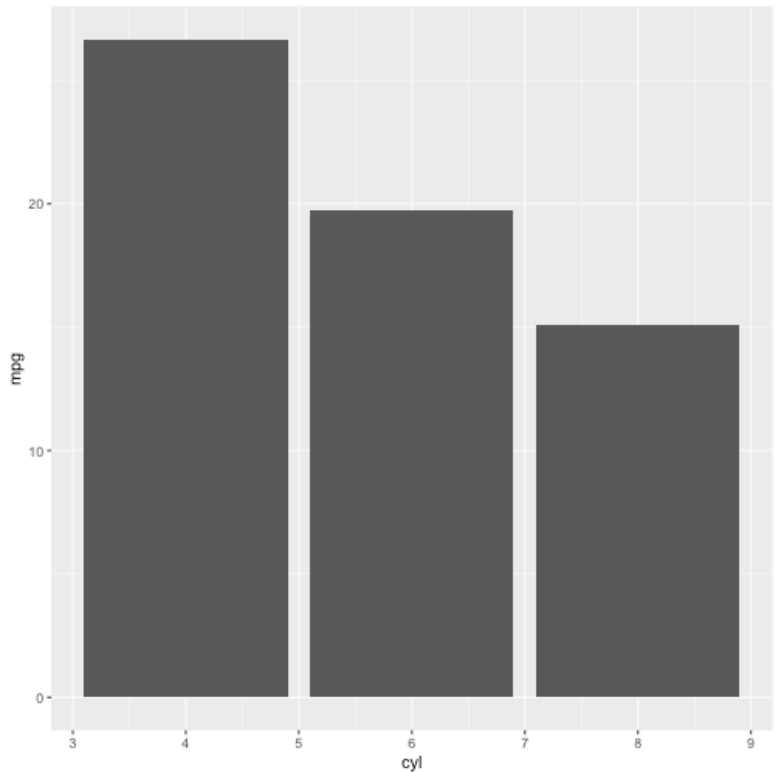
Dataset

Convert to Plot

Add Something

Fix Class Issue

```
ggplot(mtcars_summary) +  
  geom_col(  
    aes(  
      x = cyl,  
      y = mpg  
    )  
  )  
)
```



Bar Plot — Step-by-Step

Dataset

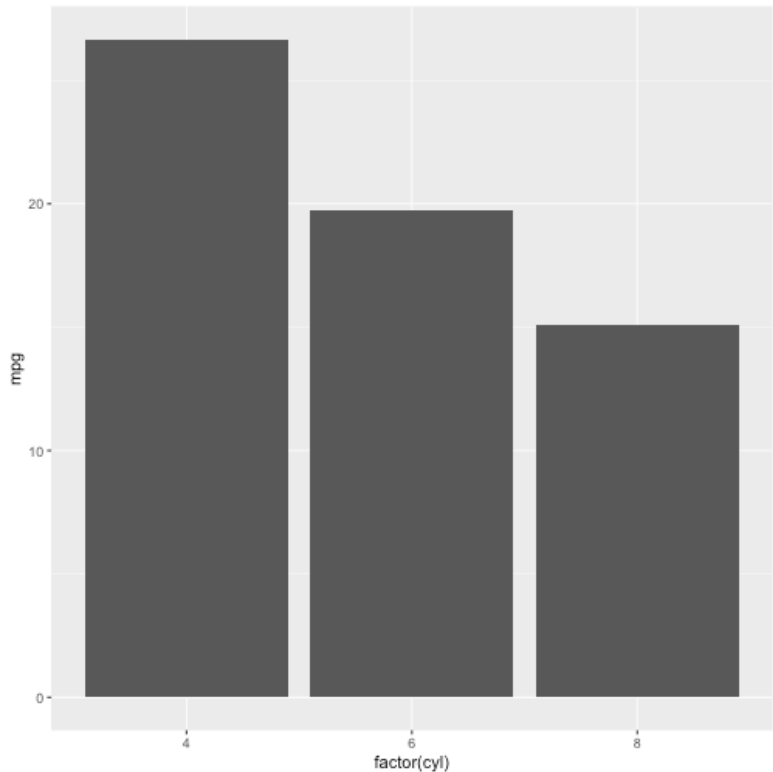
Convert to Plot

Add Something

Fix Class Issue

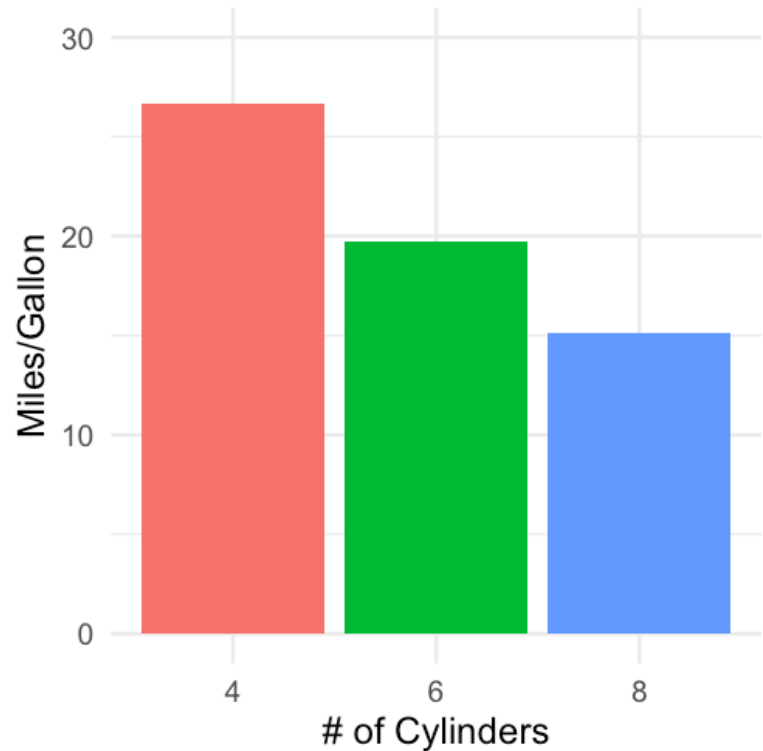
```
ggplot(mtcars_summary) +  
  geom_col(  
    aes(  
      x = factor(cyl),  
      y = mpg  
    )  
  )  
)
```

`cyl` categorizes cars by number of cylinders. Although the values are numbers, it is a **categorical** variable. We communicate this to `ggplot()` using the `factor()` function.



Bar Plot — Make It Better

```
ggplot(mtcars_summary) +  
  geom_col(  
    aes(  
      x = factor(cyl),  
      y = mpg,  
      fill = factor(cyl)  
    )  
  ) +  
  xlab("# of Cylinders") +  
  ylab("Miles/Gallon") +  
  scale_y_continuous(  
    limits = c(0, 30)  
  ) +  
  theme_minimal(base_size = 24) +  
  theme(  
    legend.position = "none"  
  )
```



Plot Standards

1. Your plot should be **properly labeled**:

- The plot should have a title describing its content
- Axes should be labeled
- Legend (if any) should have a title and labels

2. Your plot should be **properly formatted**:

- Axis dimensions should be appropriate. What is appropriate varies depending on context, but usually you should aim to fill the plot space with data
- Text size should be large enough for text to be legible

3. Your plot should be **self contained**. People should be able to understand your plot and its data without any other context or explanatory text. That means:

- A caption note that includes data source and any important data construction notes
- Title and subtitle that deliver the plot's *message*

Data Visualization — Descriptive Statistics — Tables

Descriptive Statistics Tables

Thankfully, not every RA position requires academic-standard tables or use of LaTeX.

It is still useful, however, to be able to communicate descriptive statistics about data.

There are countless R packages to help do this. Today, we're looking at the `gt` package. It's simple to use and it's very easy to create good-looking tables using it.

`gt` exports into .png, .pdf, or .html. You can add interactive elements, plots within columns.

GEWE El Salvador Baseline					
Module L -- Female HoH -- Time Use -- Weekdays					
	Mean (SD)	Median (Q1, Q3)	Min - Max	# Obs (% Group)	# in Group
Daily Time Spent (Hours)					
Agriculture (Household)	0.24 (0.81)	0 (0, 0)	0 - 10	1,275 (100%)	1,275
Childcare	0.97 (1.49)	0 (0, 1.6)	0 - 14.25	1,275 (100%)	1,275
Chores	4.76 (2.54)	4.58 (3, 6.48)	0 - 13.75	1,275 (100%)	1,275
Collecting Water	0.17 (0.48)	0 (0, 0)	0 - 4.25	1,275 (100%)	1,275
Collecting Wood	0.2 (0.54)	0 (0, 0)	0 - 4.42	1,275 (100%)	1,275
Eating	2 (1.06)	1.75 (1.29, 2.5)	0 - 9.25	1,275 (100%)	1,275
Leisure/Religion	2.39 (2.04)	2 (0.75, 3.5)	0 - 14.5	1,275 (100%)	1,275

Descriptive Statistics Table — Step-by-Step

We will mainly use the example in the script for this. To summarize, the steps are:

- Create a dataset you want to export
- Run the dataset through the `gt()` function to create a gt object
- Customize the table using functions from the `gt` package (see online for further things you can do). Examples of what you can do include:
 - Modify column names — `cols_label()`
 - Modify borders — `tab_style()`, `cell_borders()`
 - Add colors conditional on cell value — `data_color()`
 - Add title/subtitle — `tab_header()`
- Export the table using `gtsave()`

Data Visualization — Simple Regression Table

Regression Tables

Regression tables are very common in economic/policy analysis.

They're very simple to create using R and a software called **LateX** (pronounced latek).

Unless you're getting into academic research, you don't need to know how to properly use LaTeX. Just enough to:

- Export the LaTeX script from R
- Copy/paste it into a LaTeX-reading software, e.g. Overleaf
- Export the pdf or png to share

Predicted Consumption per Capita (2019 PPP USD)

	Any Treatment vs. Control (1)	Women Working Treatment vs. Any Treatment (2)
Any Treatment	12.049** (5.330)	12.155* (6.600)
Women Working Treatment		-0.222 (8.463)
Baseline Control	0.249** (0.101)	0.249** (0.101)
Constant	22.788*** (3.483)	22.791*** (3.489)
Control Mean	27.91	27.91
Observations	761	761
R ²	0.028	0.028
Adjusted R ²	0.025	0.024
Residual Std. Error	44.983 (df = 758)	45.013 (df = 757)
F Statistic	10.925*** (df = 2; 758)	7.275*** (df = 3; 757)

Note:

*p<0.1; **p<0.05; ***p<0.01

Regression Table — Step by Step

Run Regression in R

Convert to Exportable Table

```
# Simplest regression format in R
```

```
reg_example ← lm(  
  outcome_variable ~ independent_variable + control_variables,  
  data = dataset  
)
```

```
# Observe results
```

```
reg_example %>% summary()
```

Regression Table — Step by Step

Run Regression in R

Convert to Exportable Table

Simply do one of these!

```
reg_example_ht ← reg_example %>%  
  huxtable::huxreg()
```

OR

```
reg_example_sg ← reg_example %>%  
  stargazer::stargazer() # Many options to make prettier
```

Regression Table — Step by Step

Export Huxtable Table

Export Stargazer Table

Some simple options for the Huxtable table:

```
huxtable::quick_latex(  
  reg_example_ht,  
  file = "filepath/filepath/filepath/reg_example_ht.tex"  
)  
  
huxtable::quick_pdf(  
  reg_example_ht,  
  file = "filepath/filepath/filepath/reg_example_ht.pdf"  
)  
  
huxtable::quick_html(  
  reg_example_ht,  
  file = "filepath/filepath/filepath/reg_example_ht.html"  
)
```

Regression Table — Step by Step

Export Huxtable Table

Export Stargazer Table

You can export a LaTeX script using the 'writeLines' function

```
writeLines(  
  reg_example_sg,  
  "filepath/filepath/filepath/reg_example_sg.tex"  
)
```

To visualize your table, the easiest solution is to:

- Create a free Overleaf account on overleaf.com
- Open a new document
- Copy/paste your .tex output in between the `begin{document}` and `end{document}` lines
- Click compile and then save!

You can also install the `tinytex` package and use `pdftolatex` to save a PDF file.

Practical Exercise — Using the World Values Survey Dataset

World Values Survey

Background

"The survey, which started in 1981, seeks to use the most rigorous, high-quality research designs in each country. The WVS consists of nationally representative surveys conducted in almost 100 countries which contain almost 90 percent of the world's population, using a common questionnaire. [...] WVS seeks to help scientists and policy makers understand changes in the beliefs, values and motivations of people throughout the world."

Survey Contents

- Social values, attitudes & stereotypes
- Societal well-being
- Social capital, trust and organizational membership
- Economic values
- Corruption
- Migration
- Post-materialist index
- Science & technology
- Religious values
- Security
- Ethical values & norms
- Political interest and political participation
- Political culture and political regimes
- Demography

Today's practical component

1. Successfully run the code in the `session_2.R` script
2. Attempt the challenges at the bottom of the script!

Mid-Course Assignment (Due on Wednesday, October 1)

1. Find an open-source dataset. Anything will do, there is ton of data out there. If you are looking for inspiration, you can explore the [Tidy Tuesday Github repository](#).
2. Create a new folder called "**your name** R Course Assignment". Replace **your name** with your name.
3. Set up that folder for data use: - Create empty "code", "data", and "output" folders within it - Create an R Project in that folder. Refer back to last week's slides if you're not sure on how to do this.
4. Add your open-source dataset (I strongly recommend downloading or converting it to CSV format) to the "data" folder.
5. Download the "r_template.R" script from the "Course Resources" module on Canvas and place it in the "code" folder.
6. Write a script that produces and saves a plot using your open-source dataset.

Grading criteria

You will submit a .zip file of the "**your name** R Course Assignment" folder.

- Setup
 - The script's purpose should be stated at the top of the script
 - The script should load required packages for it to run
 - The script should successfully import the data used for the data visualization
- Plot
 - The plot should convey an interesting relationship between variables. For instance, you could use a scatter plot to show how two variables are positively or negatively correlated. Or you could use a bar plot to show the evolution of a variable over time.
 - The plot should meet the "plot standards" laid out in [slide 11](#).
 - The script should end with the plot being saved in the "output" folder.

To receive top grades, the script must fully run in one click without any errors.

Links

Tables

Marek Hlavac, [“stargazer: beautiful LATEX, HTML and ASCII tables from R statistical output”](#)

Thomas Mock, [“gt - a \(G\)rammar of \(T\)ables”](#)

Plots

Alicia Horsch, [“A quick introduction to ggplot2”](#)

RStudio, [RStudio Cheatsheets](#)